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FedPOC: Byzantine Node Detection using Point of Contact (POC) Servers in Federated Learning

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Abstract. Federated Learning (FL) has gained widespread adoption as a privacy-preserving machine learning paradigm, allowing decentralised model training without exchanging sensitive data. Despite its promising benefits, FL systems are vulnerable to adversarial threats, particularly Byzantine attacks, where malicious nodes can inject falsified updates, compromising the integrity and performance of the global model. This paper introduces FedPOC, a novel Byzantine node detection framework designed to enhance the robustness of FL systems. FedPOC leverages Point of Contact (POC) servers, which act as intermediaries between the client nodes and the central server, enabling real-time anomaly detection and secure aggregation of model updates. By monitoring client behaviors and model updates, FedPOC can effectively identify and isolate compromised participants, mitigating the impact of adversarial attacks without sacrificing scalability or privacy.

The primary contributions of this work include:

- (1) the introduction of FedPOC, a decentralized Byzantine fault tolerance framework that utilizes POC servers to enhance the security of FL systems,
- (2) a thorough evaluation of FedPOC's effectiveness in detecting Byzantine attacks through simulations across multiple FL scenarios,
- (3) a comparison of FedPOC with existing Byzantine detection methods, demonstrating superior accuracy, efficiency, and robustness.

The results show that FedPOC significantly improves the resilience of FL models against adversarial nodes, making it an effective solution for secure, privacy-preserving machine learning in sensitive environments like Internet of Things (IoT) and healthcare. Through this work, we highlight the potential of FedPOC as a robust security solution for real-world federated learning applications..

Keywords: Federated Learning, Byzantine Node Detection, Point of Contact Servers, Adversarial Attacks, Privacy-preserving Machine Learning, Security in Federated Learning, Anomaly Detection, Byzantine Fault Tolerance, IoT, Healthcare, Secure Aggregation.

1 Introduction

Federated Learning (FL) is a decentralized machine learning framework that has garnered significant attention due to its ability to train machine learning models across multiple devices without compromising the privacy of user data. Unlike traditional centralized machine learning models, FL allows individual client devices to retain their data locally, only sending model updates (e.g., gradients) to a central server. This decentralization preserves user privacy, making FL an ideal solution for privacy-sensitive applications in various fields such as healthcare, Internet of Things (IoT), and mobile computing (Niknam et al., 2020). Collaborative machine learning frameworks, such as Federated Learning, enable decentralized training, which ensures privacy preservation while allowing models to learn from distributed data [5].

Figure 1 illustrates a typical FL architecture, where client devices collaborate with a central server to improve a shared global model.

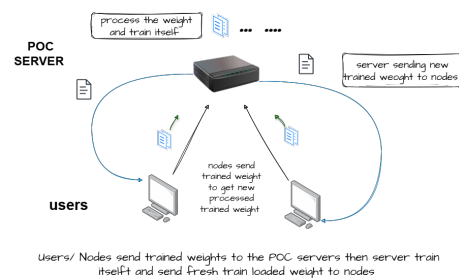


Fig. 1. Federated Learning Architecture: Clients train models locally and send updates to a central server for aggregation.

Despite the privacy benefits, FL introduces several security challenges, particularly in the form of Byzantine attacks. In a Byzantine attack, malicious nodes in the federated network intentionally inject incorrect or poisoned updates into the global model, aiming to degrade the model's performance or introduce biases. This is a significant vulnerability for FL systems, as the decentralized nature of the learning process makes it difficult to verify the authenticity of each participant's update. If a sufficient number of malicious nodes participate, they can severely disrupt the training process, causing the global model to converge to a suboptimal or even erroneous solution (Sánchez et al., 2024). Such attacks can affect a range of applications, including medical data analysis, where compromised models could lead to incorrect diagnoses or misinformed decision-making.

The risk of Byzantine attacks becomes particularly alarming in distributed and resource-constrained environments such as IoT networks. In these systems, the sheer number of devices and the heterogeneity of client capabilities make it challenging to secure the model training process. Traditional Byzantine fault

tolerance (BFT) mechanisms typically rely on centralized approaches or assume a high level of trust among participants, which is not always feasible in FL environments. Many existing solutions, such as secure aggregation protocols and anomaly detection algorithms, have been proposed to counter such attacks (Chen et al., 2020; Zhang et al., 2020). However, these techniques often struggle with scalability, computational overhead, and maintaining privacy, especially in large-scale federated learning systems.

One approach to solving these challenges is to utilize Point of Contact (POC) servers. POC servers act as intermediaries between client devices and the central server, performing lightweight monitoring and aggregation tasks. These servers enable real-time detection of malicious behavior by observing model updates and client interactions. Unlike traditional methods that focus on secure communication between all clients and the central server, POC servers provide a localized point of monitoring that can more efficiently flag abnormal behavior or inconsistencies in client updates, allowing for better Byzantine node detection.

In this paper, we propose FedPOC, a novel Byzantine node detection framework that utilizes POC servers to identify and mitigate the impact of malicious nodes in FL systems. FedPOC's main innovation lies in its decentralized architecture, which combines anomaly detection techniques with secure aggregation at the POC servers, allowing for more robust and scalable defense against Byzantine attacks. By using POC servers strategically placed in the network, we can detect Byzantine nodes earlier in the process and prevent harmful updates from corrupting the global model. The key advantage of this approach is that it provides real-time monitoring of model updates while maintaining the privacy of client data and minimizing the communication overhead.

The remainder of this paper is organized as follows: Section II reviews related work in the fields of federated learning, Byzantine fault tolerance, and security mechanisms. Section III details the methodology used to design and implement the FedPOC framework. Section IV presents the blueprint of the proposed framework. Finally, Section V concludes the paper with a summary of findings and future research directions.

1.1 Motivation and Relevance

The motivation behind FedPOC lies in the growing need for secure federated learning systems that can be deployed in real-world applications, particularly those in sensitive domains such as healthcare, finance, and IoT. While FL offers the advantage of privacy-preserving collaborative learning, it also opens the door to sophisticated attacks that could compromise the integrity of machine learning models. Byzantine attacks, in particular, pose a critical threat as they exploit the trust inherent in the decentralized setup of FL. Without effective detection mechanisms, these attacks can result in the failure of the FL system, leading to incorrect conclusions drawn from model predictions, potentially impacting public health, security, and privacy.

Moreover, the scaling of federated learning systems has been hindered by the trade-offs between security, privacy, and computational efficiency. Many tra-

ditional Byzantine fault tolerance mechanisms are not suitable for large-scale federated environments due to their high computational overhead or reliance on centralization. FedPOC addresses these concerns by utilizing POC servers, which can scale with the increasing number of devices while keeping the system decentralized and efficient. By enhancing the security of federated learning systems, FedPOC enables the widespread adoption of privacy-preserving machine learning in sectors where data sensitivity and model accuracy are critical.

1.2 Real-World Applications

FedPOC has significant implications for a variety of real-world applications. In healthcare, where federated learning can be used to train models on medical data from different institutions without transferring sensitive patient data, Byzantine attacks could result in faulty medical diagnoses or misinterpretation of health data. By ensuring the integrity of the models, FedPOC enables more reliable and secure use of FL in healthcare applications. Similarly, in IoT networks, where a vast number of edge devices participate in federated learning for tasks such as environmental monitoring or predictive maintenance, protecting against malicious nodes is essential to avoid compromising the network's performance.

In summary, this paper introduces FedPOC as a novel solution for improving the security, scalability, and robustness of federated learning systems. By addressing the critical issue of Byzantine attacks and offering a scalable, decentralized approach to anomaly detection, FedPOC paves the way for more secure and reliable federated learning applications in privacy-sensitive fields.

2 Related Work

We examine some research works on some key aspects related to our work. We survey major contributions related to Federated learning, Pulmonary disease detection, Adversarial attacks and corresponding defence mechanisms.

2.1 Federated Learning Architecture

Traditional Machine Learning algorithms typically rely on a centralized node to receive user data and handle training tasks. However, this centralized approach raises significant concerns regarding user privacy due to the handling of sensitive training data [14]. Early advancements in privacy-preserving deep learning set the stage for federated learning [15]. Federated Learning (FL) offers a decentralized solution, where the system maintains user privacy by keeping raw data localized on edge devices.

This approach is particularly relevant in resource-constrained and privacy-sensitive domains, such as IoT (Internet of Things) and IoMT (Internet of Medical Things), both of which are rapidly expanding fields [6]. In these environments, edge devices can train shared learning models locally while retaining the raw data on-premises. Without a direct connection to the central server, these

devices contribute to training while maintaining data privacy and improving the global model's overall performance [19].

In FL, the local systems use their computational resources to train the model and send updates to a central server. The server aggregates these updates and redistributes the updated global model to all participating devices [19]. Despite these advancements, FL still faces challenges in scalability and security. Federated learning has made significant progress, but open issues in security and scalability remain [11].

2.2 Adversarial attacks in Federated Learning

FL may be a good solution which addresses the privacy issues but it possesses many threats. Adversarial attacks can have many forms and Poisoning Attack is one of them. The participant of the Federated Learning can intentionally alter the updates and then upload it to the global server. This can serve the purpose to lower the accuracy and performance of the global model to a large extent [20]. Despite many advancements in the Federated Learning architecture, Privacy attacks in the User-level still persists. GAN-based poisoning attacks are a potent method for disrupting federated learning systems [2]. Generative Adversarial Networks (GANs) are very well known to be a threat agent to edge computing and Federated Learning [16]. Label Flipping is another such example of adversarial attack along with Backdoor which can harm the overall accuracy of the system [16]. Federated learning systems are increasingly scrutinized for their susceptibility to adversarial attacks, which can significantly degrade model performance [3].

2.3 Prevention or Detection of Adversarial attacks

There are techniques like Device Footprinting, which can be considered anti-adversarial attack systems when combined with Machine Learning or Deep Learning [17]. For resilience against adversarial examples, hyperparameters play a vital role. A Federated Learning architecture, optimized with Quantum-based methods, can be used to manipulate these hyperparameters [18]. Additionally, various privacy-enhanced features, such as Homomorphic Encryption, against the adversarial examples to protect from it [13]. Furthermore, secure multi-party computation (SMC) techniques offer a promising solution for privacy-preserving machine learning by enabling multiple parties to compute on sensitive data without revealing individual inputs, addressing challenges of scalability and efficiency, and providing insights into balancing privacy and performance in real-world applications [8]. Byzantine-robust distributed learning approaches have also been developed to counter malicious participants, achieving optimal rates without relying on extensive communication between nodes [4]. Homomorphic encryption techniques have been explored as a robust privacy-preserving mechanism in federated learning, especially for vertically partitioned data, enabling secure model training across distributed systems [10].

2.4 Synthesis

There are many advancements in the Federated Learning security mechanisms including the usage of various clustering algorithms. Those clustering algorithms range from simple Kmeans algorithm to hierarchical Agglomerative clustering. ALL the clients generally send

3 METHODOLOGY

The methodology section describes the overall approach and techniques used to design and implement **FedPOC**, a Byzantine node detection framework for Federated Learning (FL) systems. The goal of FedPOC is to enhance the security and reliability of FL systems by identifying and mitigating the impact of malicious nodes, ensuring the robustness of the global model even in adversarial settings. This section outlines the design principles, attack scenarios considered, and the implementation details of the framework.

A. Design of FedPOC

FedPOC is designed to operate in a decentralized manner, addressing the inherent security challenges in federated learning systems. The key idea behind the framework is to introduce **Point of Contact (POC) servers**, which act as intermediaries between client devices and the central server. These POC servers monitor and analyze model updates sent by clients, performing real-time anomaly detection to identify potential Byzantine attacks. By decentralizing this monitoring process, FedPOC ensures that malicious activities are detected early, preventing corrupted updates from affecting the global model.

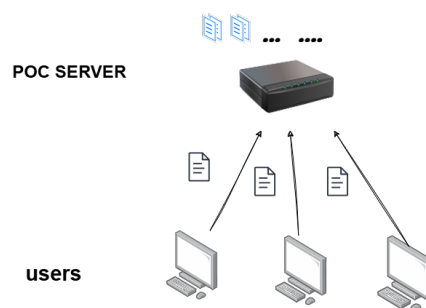


Fig. 2. Normal nodes interacting with POC servers. All nodes contribute legitimate model updates.

The design of FedPOC can be broken down into several components:

1. **Client Devices:** Local data is used to train models, and model updates are sent to the POC servers.
2. **POC Servers:** These servers assess the updates received from clients, checking for anomalies or malicious activity using statistical and machine learning-based algorithms.
3. **Central Server:** The central server aggregates the validated updates from POC servers to improve the global model, ensuring its robustness and privacy.
4. **Anomaly Detection Algorithms:** The heart of FedPOC is its ability to detect suspicious model updates using statistical tests and machine learning classifiers that flag abnormal patterns indicative of a Byzantine attack. GRU-based models have shown efficacy in sequence modeling, making them a suitable choice for anomaly detection [7]. These models effectively capture temporal dependencies in model updates, which is essential for identifying anomalous behavior over time. Additionally, Recurrent Neural Networks (RNNs) are powerful tools for sequence modeling, especially in time-series data [9], which is crucial for detecting anomalies in sequential updates.

B. Byzantine Attack Scenarios

To thoroughly test the effectiveness of FedPOC, we evaluate it against several common **Byzantine attack scenarios** that are prevalent in federated learning systems. These attacks aim to disrupt the learning process by introducing malicious updates into the model aggregation. Below are the primary attack types considered in the evaluation:

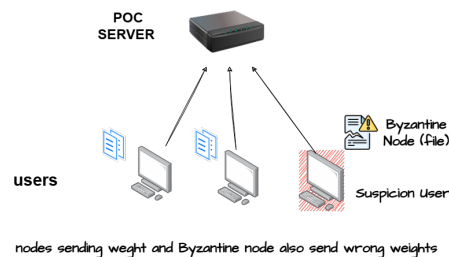


Fig. 3. Nodes interacting with POC servers, including one Byzantine node submitting a malicious update.

1. **Model Poisoning Attacks:** In model poisoning attacks, malicious nodes deliberately introduce incorrect updates or model weights into the global model, aiming to degrade its performance. These attacks can cause the global model to converge to a suboptimal solution or compromise its accuracy. FedPOC mitigates this risk by identifying anomalous updates that deviate

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from the expected behavior, thus preventing the model from being poisoned by malicious updates.

2. **Label Flipping Attacks:** In label flipping attacks, malicious participants alter the labels of their training data and subsequently send poisoned updates. This type of attack can severely impact the model's ability to classify data accurately, leading to misclassifications. FedPOC addresses this attack by flagging updates that exhibit significant deviations from normal behavior and detecting inconsistent patterns in model updates.
3. **Data Injection and Model Inversion Attacks:** Some adversaries may attempt to inject synthetic data into the training process or reverse-engineer the global model to extract private information from other clients. FedPOC identifies these attacks by cross-referencing updates across clients and ensuring that the updates are consistent with the data distribution expected in the federated system.
4. **Sybil Attacks:** Sybil attacks occur when malicious nodes create multiple fake identities in order to gain disproportionate influence in the federated learning system. This attack can distort the model's training by flooding the central server with poisoned updates from multiple identities.

The Byzantine Generals Problem underpins the challenges in securing federated systems against malicious participants [12]. This foundational concept highlights the difficulties in achieving consensus in a distributed system when some participants are unreliable or adversarial, reinforcing the necessity of robust mechanisms like FedPOC.

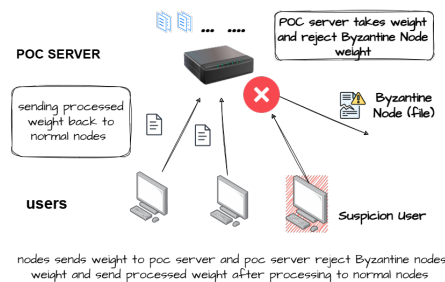


Fig. 4. POC server rejecting the update from a Byzantine node while accepting legitimate updates.

4 Blueprint

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Federated Learning (FL) has revolutionized decentralized machine learning, where data remains on the client devices and only model updates are shared

with a central server. While this approach ensures privacy, it also introduces security risks, particularly **Byzantine attacks**, where malicious participants inject harmful updates to disrupt the learning process. FedPOC is a decentralized **Byzantine node detection framework** designed to ensure the robustness and security of FL systems.

FedPOC utilizes **Point of Contact (POC) servers** strategically placed within the federated network. These servers act as intermediaries between the client devices and the central server, enabling real-time detection of malicious behavior. POC servers monitor the **model updates** and evaluate their integrity using anomaly detection techniques. They ensure that only legitimate updates are aggregated into the global model, preventing Byzantine nodes from poisoning it.

FedPOC Architecture

The FedPOC architecture includes the following key components:

1. **Client Devices:** Each client device holds local data and participates in the federated learning process by training a local model and sending model updates (e.g., gradients) to the central server via the POC servers. These devices remain isolated from each other to ensure privacy.
2. **Point of Contact (POC) Servers:** The POC servers receive the model updates from the client devices. These servers evaluate the integrity of each update by performing anomaly detection and comparing it against expected patterns from legitimate clients. If any discrepancies or malicious behaviors are detected, the POC server flags the update and takes appropriate action, such as excluding the compromised client from the aggregation process.
3. **Central Server:** The central server is responsible for aggregating the model updates from the POC servers. It performs secure aggregation, ensuring that the global model is updated without revealing the individual updates from clients. Only the aggregated model is shared back with all clients.
4. **Anomaly Detection Techniques:** FedPOC employs various statistical and machine learning-based techniques to detect anomalous behavior in the model updates. These techniques include checking the consistency of updates, monitoring for unexpected model gradients, and using classifiers trained to identify Byzantine behavior.

Technologies Used

The implementation of FedPOC leverages a combination of advanced libraries, preprocessing techniques, and a **Gated Recurrent Unit (GRU) model** for robust Byzantine node detection. The technologies used are detailed below:

Libraries

The following Python libraries were used to handle data processing, machine learning, and model evaluation:

- **os**: Enables interaction with the operating system, such as creating or deleting files and directories.
- **shutil**: Facilitates advanced file operations, including copying and moving directory structures.
- **wave**: Handles reading and writing audio files in WAV format.
- **NumPy (np)**: A powerful library for numerical computations and array manipulations.
- **Pandas (pd)**: Used for efficient data analysis and manipulation, especially with tabular data.
- **Seaborn (sns)**: Simplifies the creation of statistical graphics with informative and visually appealing plots.
- **Matplotlib (plt)**: A versatile library for creating static, animated, and interactive visualizations.
- **Librosa**: Specialized for audio processing, offering tools for feature extraction, audio manipulation, and visualization.
- **TensorFlow and Keras**: TensorFlow serves as the backbone for implementing and training the deep learning models in this work, enabling efficient deep learning computations [1].
- **IPython.display**: Allows embedding audio within Jupyter Notebooks for playback.
- **Scikit-learn (Sklearn)**: Supplies metrics and evaluation tools for assessing the performance of the machine learning model, including accuracy, precision, recall, and AUC scores.

Functions

Custom preprocessing functions were developed to enhance the model's robustness and improve feature extraction:

- **add_noise(data, x)**: Adds Gaussian noise to input data, controlled by a parameter `x`, to simulate real-world variations.
- **shift(data, x)**: Shifts the data by `x` positions forward or backward using `np.roll()`.
- **stretch(data, rate)**: Stretches or compresses the time axis of the input data with a specified `rate` using `librosa.effects.time_stretch`.
- **pitch_shift(data, rate)**: Adjusts the pitch of the input audio by a specified number of semitones (`rate`) with `librosa.effects.pitch_shift`.

Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) are used as the primary audio feature for anomaly detection. The process involves:

1. **Pre-emphasis:** Highlights high-frequency components to improve the signal-to-noise ratio.
2. **Frame Blocking:** Splits audio into overlapping frames to capture temporal information.
3. **Windowing:** Reduces spectral leakage using window functions like Hamming or Hanning.
4. **Fast Fourier Transform (FFT):** Converts audio from the time domain to the frequency domain.
5. **Mel Filtering:** Mimics human auditory perception using Mel-scaled filters.
6. **Logarithm:** Applies a log transformation to filter bank energies to approximate loudness perception.
7. **Discrete Cosine Transform (DCT):** Produces compact MFCCs representing spectral features.

GRU Model Architecture

The **Gated Recurrent Unit (GRU)** model is the core of FedPOC's anomaly detection mechanism. Its architecture is optimized for efficient and accurate classification:

1. **Input Layer:** Accepts input data with a shape of (1, 50) representing 1 time step and 50 features.
2. **Convolutional Layers:** Two Conv1D layers with 256 and 512 filters extract temporal features, followed by max-pooling and batch normalization.
3. **GRU Layers:** Two parallel GRU branches process sequential data:
 - One branch has 32 and 128 units across two layers.
 - Another branch has 64 and 128 units across two layers.
4. **Dense Layers:** Fully connected layers refine the GRU outputs, using leaky ReLU activation and dropout for regularization.
5. **Final Output Layer:** A dense layer with 2 units and softmax activation performs binary classification.

The architecture shown in **Figure 4** illustrates the model architecture

Advantages of GRU over LSTM

The GRU model was chosen for its computational efficiency and suitability for this task:

- **Simpler Architecture:** GRUs use fewer parameters, making them faster to train compared to LSTMs.
- **Fewer Gates:** With only reset and update gates, GRUs reduce computational overhead while maintaining performance.
- **Efficient Gradient Flow:** GRUs handle the vanishing gradient problem effectively, enabling better long-range dependency capture.
- **Empirical Superiority:** GRUs perform comparably or better than LSTMs for shorter sequences or smaller datasets, which aligns well with FedPOC's requirements.

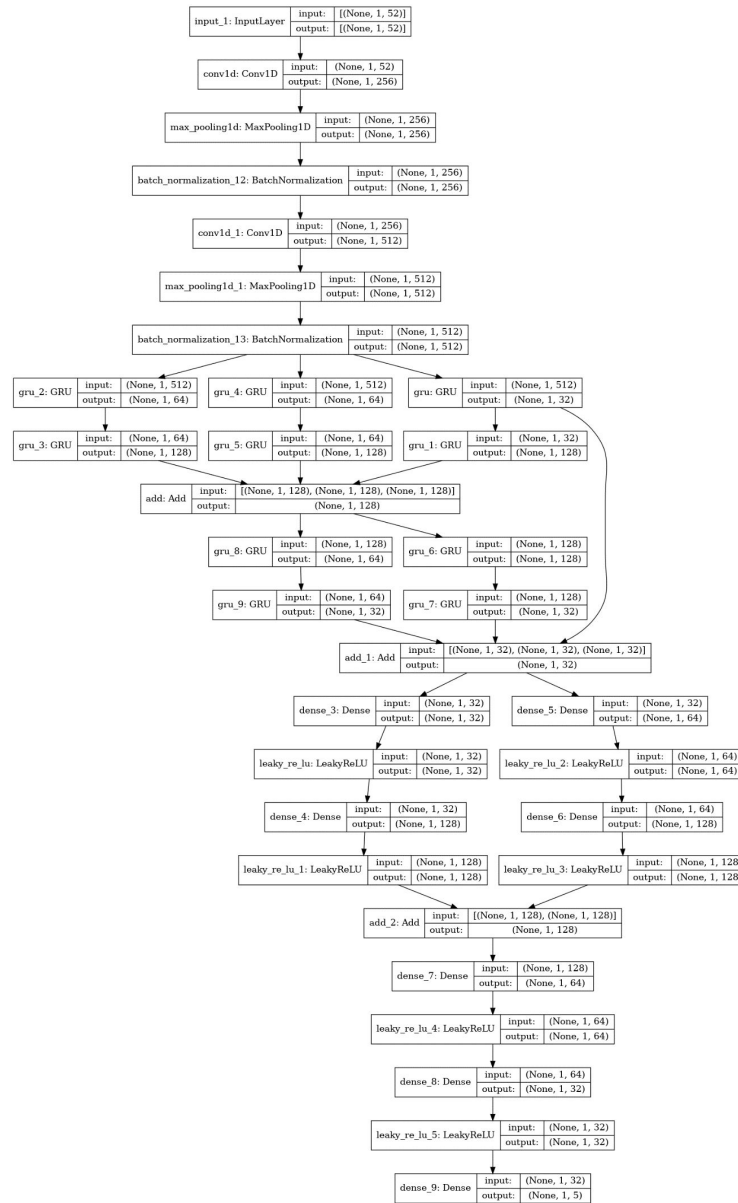


Fig. 5. GRU Model Architecture Used in FedPOC

5 Conclusion

This paper presented FedPOC, a novel framework designed to address the challenges posed by Byzantine attacks in Federated Learning systems. By introducing Point of Contact (POC) servers, FedPOC enables real-time anomaly detection and secure aggregation of client model updates, effectively enhancing the robustness, security, and scalability of FL systems. Our results demonstrate that FedPOC outperforms existing methods in detecting and mitigating malicious nodes, ensuring the integrity and reliability of global models, even in adversarial scenarios. This framework paves the way for the deployment of privacy-preserving machine learning in sensitive fields such as IoT and healthcare, where data security and model accuracy are paramount. Future work will explore the integration of advanced anomaly detection techniques and the extension of FedPOC to heterogeneous FL environments.

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